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Optimal Trajectory Planning for Multiphase Lunar Landing ⋆

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Abstract: Lunar landing problem is formulated as an optimal control problem and has been solved by Legendre Pseudospectral method. The landing problem is split into various stages to take into account various mission constraints. Based on these constraints, the problem is split into three stages−Rough Braking, Attitude Hold and Retargeting . Legendre Pseudospectral method is used for solving this multiphase lunar landing problem. The idea of this method is to discretize the trajectory optimization as a nonlinear programming problem. Legendre Pseudospectral method was applied to approximate the state and the state differential equations as algebraic constraints. Optimal solutions of the above problem are those control variables which dynamically satisfy all mission constraints. The cost function considered is minimum fuel satisfying all mission constraints envisaged. The method guarantees a minimum fuel solution which satisfies terminal mission constraints in each stage as part of the formulation. The mission constraints considered in studying this problem are desired altitude, attitude and velocity while reaching the desired landing site from a specified perilune height. At touchdown, the lander orientation should be vertical favoring landing leg of the Lander to touch the Moons surface as desired. The commanded acceleration should be within the limit of the main engine thruster capability (saturation limit) which enforces maximum and minimum thrust constraints.

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1. INTRODUCTION

Precise landing of lunar modules has been an interesting research topic ever since Man landed on the Moon and has realized that as a base, Moon provides great many opportunities to reach outer planets with ease. After the success of Chandrayaan-I (INDIA's first Moon Orbiter), Moon has been seriously considered as a base for extended missions for ISRO. In Chandrayaan-I the impact probe is released from the orbiter to impact on the moon surface. Extended mission Chandrayaan-IIs plan is to deboost the lander craft from the orbiter's composite module for soft landing on safe landing site. In order to achieve this requirement, it is necessary to develop skills to soft land the supplies and sensitive instruments on to the Moon precisely and accurately. The major technological aspects of a soft landing lunar mission are to assure precise landing, soft touch down and retargeting for hazard avoidance.

During atmospheric entry and landing, there has been unique features that are to be encountered and conquered for efficient landing. Entry into planets with atmosphere can be done by techniques like aero braking, aero capture, ballistic entry, skip entry, gliding entry etc. However, when entry is attempted into celestial bodies without atmosphere, not only the advantage one gets in reducing the velocity using the drag force is lost but also entirely the reduction should take place using the limited fuel one carries onboard. Therefore, a careful plan and execution of the burns is important in landing the modules in celestial bodies without atmosphere.

The space probe orbiting the Moon needs to land on the surface satisfying all mission constraints while complying vehicle limitation and capabilities. For this, guidance trajectory is generated by treating space probe as a point mass and the entire problem as an optimization problem.

The various phases for a lunar module to land on the Moon are provided in the next section followed by a review of some of the published works.

Hawkins (2005) presented a detailed investigation of the constrained trajectory optimization of the Moon landing problem. Assumptions include a spherical, homogeneous, and non-rotating Moon, a vehicle with one throttlable fixed main engine with a constant exhaust velocity, and a purely Newtonian gravity model. The trajectory was examined from an initial circular parking orbit of 40 km. Constraints were progressively added to mitigate the operationally undesirable characteristics of the obtained solutions. The resulting trajectories were analyzed based on state and control histories, effect on fuel usages, operational feasibility. Kozynchenko (2010) has addressed

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a problem of expanding the attainable downrange area for a lunar landing module while guiding to a specified point. Both the conventional boundary conditions and the restrictions on fuel consumption and throttling coefficient are taken into account. More specifically, the purpose of algorithm development is to provide the equality of the available thrust margins over the maximal remaining part of a descent trajectory at the braking phase. Three kinds of predictive algorithms are considered: the first one having a constant thrust over a trajectory and not providing the equal thrust margins, the second and third have a stepped thrust profiles and a jump distance as a control. The first and second, so called planning algorithms are able to generate the complete control profiles at the beginning of the braking phase, whereas the third one, gradual or spreading-over-path, computes the control profile in real-time fashion gradually when moving over the trajectory.

Cho et al. (2009) in their paper studied the perilune altitude of the intermediate orbit with optimal thrust programming for minimal energy. The perilune altitude and optimal thrust programming can be expressed as a function of the radius of a parking orbit by using continuation method and co-state estimator. For designing optimal lunar landing trajectories, perilune altitude turned out to be important in reducing the total energy for the lunar landing. Therefore, the initial state of descent phase is free in this optimal control problem. Using shooting method and the continuation approach, the optimal solution under various parking orbit conditions has been found. Rekha et al. (2012) focuses on the navigation system required for the Lunar Soft Lander which is a complete, integrated, autonomous system with multiple sensor information for precise and soft landing. The challenges in the navigation system design for safe, precise, soft and autonomous landing on the Moons surface are discussed. The Navigation Guidance Control (NGC) system has to perform several functions that are critical to the soft landing. The NGC configuration during various phases of the mission is also described. The mission profile including the navigation plan from the time of lift off till landing on moons surface is briefed. The NGC sensor suite was selected catering to the functional requirements of the mission. The selection of specific sensors during various phases is also described.

A logical development of a novel, minimum-complexity guidance system for precise and soft lunar landing is presented by Vaeth and Sarles (1964) and they provide an evaluation of predominant error sensitivities. A new trajectory control technique presented in this paper allows excellent compromise between sensor requirements (e.g., Doppler and beacon tracking radar gimbaling), control system complexity and total fuel usage. Lee (2011) proposed an optimal control law for a lander to achieve a soft touchdown. For simplicity, the vehicle is modeled as a point mass and only two-dimensional planner motion is considered. In this study, a cost functional that penalizes both the terminal vertical and horizontal velocities of the vehicle as well as the fuel cost of the descent main propulsion system is considered. Park et al. (2011) address a minimum-fuel, two-dimensional trajectory optimization for a soft lunar landing from a parking orbit to a desired landing site using Legendre Pseudospectral (LPS) knotting method. Designing precise trajectories for landing at a desired site demand the landing site to be considered as the terminal constraint in the formulation. Zhou et al. (2010) considered an optimal control problem arising from the guidance of a lunar module to achieve soft landing, where the description of the system dynamics is in a three-dimensional coordinate system. Optimal guidance law for soft landing of the lunar module with the terminal attitude of the module to be within a small deviation from vertical orientation with respect to lunar surface has been proposed.

The above literature review gives an idea of various designs carried out for lunar landing missions. Research has been carried out for solving this problem as single stage using point mass model representing the lander craft. Earlier Mathavaraj et al. (2012) solved the moon landing powered descent phase using LPS method and Newton method. This paper address the advantages of direct and indirect optimization method. The moon landing has been solved as a single stage problem considering only few mission constraints.

But solving the landing problem as one stage is not realistic in nature. In practical scenario the optical control formulation should account for camera sensor processing time and its attitude requirement. Including this additional constraint with soft landing constraint, the optical control formulation is explained in this paper.

It is to be noted that only terminal phase of the lander problem is considered and the de-orbit phase and transfer-orbit phase have been tacitly assumed to be solved suitably. The equation of motion used for optimal control formulation is two dimensional. By optimal control formulation all mission constraints have been formulated and included in the design.

2. LEGENDRE PSEUDOSPECTRAL GUIDANCE SCHEME

Collocation methods use piecewise-continuous functions to approximate the state and control variables at the arbitrary subintervals. However, the LPS method uses globally interpolating Lagrange polynomials as trial functions to approximate the state variables, control variables, and state differential equations at the nodes, which are called Legendre-Gauss-Lobatto (LGL) points.

The reference time interval τ_i, τ_f can be transformed to LGL points t_0, t_N by change of variables

$$t \rightarrow 1 + 2(\tau - \tau_i)/(\tau_f - \tau_i) \tag{1}$$

This transformation from the real time domain to the LGL time domain is required because the LGL points are defined between [-1 1].So points are given by $t_l \quad l = 0, .., N$ where $t_0 = -1, t_N = 1$ and other LGL points are the zeros of the derivative of the Legendre polynomial of degree N (L_N). The number of LGL points is chosen based on the problem specification i.e. using the obtained optimal solution propagate the system dynamics to verify whether the terminal constraints are properly met. If not, increase the number of points and repeat the above procedure. But recommended number of LGL points is around 20. As stated we start by approximating the continuous state and control variables by Nth degree polynomial of the form

$$\begin{aligned} X(t) \approx X^N(t) &= \sum_{l=0}^{N} X(t_l)\rho_l(t) \\ U(t) \approx U^N(t) &= \sum_{l=0}^{N} U(t_l)\rho_l(t) \end{aligned} \tag{2}$$

Equation 2 is polynomials of Nth order which interpolate the functions at the LGL points. Lagrange polynomial used is

$$\begin{aligned} \rho_l(t) &= \frac{g(t)}{(t-t_l)\,\dot{g}(t)} \\ g(t) &= \left(1-t^2\right)\dot{L}_N(t) \\ \dot{g}(t) &= -N(N+1)\,L_N(t) \end{aligned} \tag{3}$$

which has kronecker delta property.

$$\begin{aligned} \rho_l(t) &= \frac{1}{N(N+1)\,L_N(t_l)} \times \frac{\left(1-t^2\right)\dot{L}_N(t)}{t-t_l} \\ \rho_l(t) &= \begin{cases} 1 & if \quad l=k \\ 0 & if \quad l \neq k \end{cases} \end{aligned} \tag{4}$$

From this property of it follows that

$$\begin{aligned} X^N(t_l) &= X(t_l) \\ U^N(t_l) &= U(t_l) \end{aligned}$$

To carry out the approximation of the left hand side of the state equations, we impose the condition such that the approximations above satisfy the differential equations at the LGL collocation points. To express the derivative at the collocation points we differentiate equation 2.

Evaluating we get following $(N+1)\times(N+1)$ matrix $D_{kl} = \dot{\rho}_l(t_k)$ which discretize the right hand side of the state equation.

$$D = [D_{kl}] = \begin{cases} \dfrac{L_N(t_k)}{L_N(t_l)(t_k-t_l)} & k \neq l \\ -\dfrac{N(N+1)}{4} & k = l = 0 \\ \dfrac{N(N+1)}{4} & k = l = N \\ 0 & otherwise \end{cases} \tag{5}$$

The optimal cost function considered is

$$J = \varphi\left(X(\tau_f), \tau_f\right) + \int_{\tau_0}^{\tau_f} L\left(X(\tau), U(\tau), \tau\right) d\tau \tag{6}$$

Using change of variables and Gauss-Lobatto integration rule we get

$$d\tau = \frac{(\tau_f - \tau_0)}{2} dt$$

$$\begin{aligned} J &= \varphi\left(X(t_f), t_f\right) + \frac{(\tau_f-\tau_0)}{2}\int_{-1}^{1} L dt \\ J^N &= \varphi\left(X^N, t_N\right) + \frac{(\tau_f-\tau_0)}{2}\int_{-1}^{1}\left(\sum_{k=0}^{N} L^k \rho_l(t)\, dt\right) \\ J^N &= \varphi\left(X^N, \tau_N\right) + \frac{(\tau_f-\tau_0)}{2}\sum_{k=0}^{N} L^k \int_{-1}^{1}\left[\rho_l(t)\right] dt \end{aligned}$$

So the above optimal control problem is approximated by the following nonlinear programming problem:

Cost Function

$$\begin{aligned} J^N &= \varphi\left(X^N, \tau_f\right) + \frac{(\tau_f-\tau_0)}{2}\sum_{k=0}^{N} L^k \begin{pmatrix} X(t_k), \\ U(t_k), \\ t_k \end{pmatrix} w_k \\ w_k &= \frac{2}{N(N+1)}\frac{1}{\left[L_N(t_k)\right]^2} \end{aligned} \tag{7}$$

Optimal Constraints

$$\frac{(\tau_f-\tau_0)}{2} f\left(X(t_k), U(t_k), t_k\right) - C_k = 0$$

$$\begin{aligned} g_l &\leq g\left(X(t_k), U(t_k), t_k\right) &&\leq g_u \\ \eta_{0l} &\leq g\left(X(t_0), U(t_0), t_0\right) &&\leq \eta_{0u} \\ \eta_{fl} &\leq g\left(X(t_N), U(t_N), t_N\right) &&\leq \eta_{fu} \end{aligned} \tag{8}$$

Pseudospectral knotting framework allows solving a multiphase optimal control problem In general LPS method needs more nodes for accurate solution for such multiphase problem. But pseudospectral knots allow calculating solution efficiently with fewer nodes which reduces large computational time. Pseudospectral knotting method idea is to divide the entire time domain into smaller time subdomain where the control and state variations are large and using double LGL points called knots between these subdomain to enforce the continuity of the states. Here LPS method is applied in each of this subdomain for discretizing into nonlinear programming problem. For more details on LPS method refer Fahroo and Ross (2004).

3. MULTIPHASE MOON LANDING FORMULATION

Lunar craft is de-boosted from a circular orbit of 100 km. De-boosted lander craft enters into Hohmann transfer trajectory from 100 km to 18 km. At perilune height of 18 km , probe enters into powered descent phase. Near this phase, the propulsive engines are turned on and the probe velocity is decreased in order to enable the craft to soft land on the lunar surface.

3.1 Phases in Moon Landing

There are three distinct phases that need to be performed in order to land the lunar module on to the surface of Moon. They are: (i) De-orbit Maneuver phase (ii) Transfer orbit phase or coasting phase and (iii) Powered descent phase. The parking orbit of the lunar spacecraft is planned to be 100 Km circular for this study. Otherwise, the method developed is general in nature and can be applied to other types of orbit as well.

Deorbit maneuver This de-orbit phase is essential to bring the vehicle from the parking orbit of 100 km to a suitable perigee height, say 18 km. Thruster firing is executed based on the principles of Hohmann transfer between the parking orbit and the descent orbit.

Transfer orbit This phase consists transition of lunar module from the parking orbit to descent orbit. Note this maneuver is carried out in such a way that the desired initial condition of the powered descent phase has been achieved in terms of position, velocity, and attitude. This is one of the main requirements in order to land the module precisely to a landing site; otherwise retargeting and out of plane maneuver will result in precious fuel expenditure.

Powered Descent This phase starts from the desired terminal condition of the transfer orbit. This phase is split into three stages due to mission constraint. First stage is named as Rough Braking Phase and its objective is to break the landers velocity suitably before a specified altitude for camera imaging. Hazard detection camera is used for identifying hazard free landing site. Stage 2 named as Attitude Hold Phase as its name suggest the attitude of the lander is withheld at specified orientation for better imaging of the lunar surface. After imaging, the onboard navigation system processes it to determine the position of the lander with respect to hazard free site. The goal of the stage 3 i.e. Retargeting Phase is guiding lander to identified safe site, reducing its kinetic energy and to orient the lander vertical suitable for landing.

In this paper, powered descent phase is analyzed assuming that deorbit maneuver phase, transfer orbit phase is carried out resulting in assumed initial condition for descent phase.

3.2 System Dynamics

In order to describe the motion of a dynamic system it is necessary to define a suitable coordinate system and formulate equations for the motion in accordance with physical laws governing the system.

Considering the vehicle motion's as a point pass flying inside a planetary atmosphere, the equation of motion is defined as

$$\begin{aligned} \dot{r} &= w \\ \dot{\theta} &= \frac{u}{r} \\ \dot{w} &= \frac{T\sin\beta}{m} - \frac{\mu}{r^2} + \frac{u^2}{r} \\ \dot{u} &= \frac{T\cos\beta}{m} - \frac{uw}{r} \\ \dot{m} &= -\frac{T}{I_{sp}g} \end{aligned} \tag{9}$$

The dynamics represent the two dimensional motion of the point mass model. In equation 9, r represents the radial distance from the center of the moon, θ represents the longitude, u, w represents the tangential velocity and vertical velocity components respectively and m represents the instantaneous mass of the lander. Also μ is the gravitational parameter of planet considered, g gravity of the planet and I_{sp} is specific impulse of the engine considered for simulation. The control parameters considered are thrust T and its orientation β. The state and control vector considered as

$$\begin{aligned} X &= [\, r \;\; \theta \;\; u \;\; w \;\; m \,]^T \\ U &= [\, T \;\; \beta \,]^T \end{aligned}$$

For further details one can refer to Vinh et al. (1980)

3.3 Dynamical Constraints

For the optimization algorithm, it is better to have all variables in the non-dimensional domain. So we do the normalization using basic variables like lander's initial distance (r_i) from the center of moon, radius of the moon (r_m) and lander's initial mass (m_i).

$$\begin{aligned} \|V\|_n &= \sqrt{\mu/r_i} \\ \|d\|_n &= r_m \\ \|m\|_n &= m_i \end{aligned}$$

A lunar craft has to be minimizing fuel consumption if it is to carry a heavy payload of mission equipment for lunar explorations. So the cost function for the optimal control problem is selected as

$$J = \int_{t_0}^{t_f} \left(\left(\frac{T\cos\beta}{m} \right)^2 + \left(\frac{T\sin\beta}{m} \right)^2 \right) dt \tag{10}$$

The dynamical equation used for all phase is formulated into non-linear programming problem as given below

$$\begin{aligned} \sum_{l=0}^{N} D_{kl} r\,(t_l) - \frac{(\tau_f - \tau_i)}{2} w|_k &= 0 \\ \sum_{l=0}^{N} D_{kl} \theta\,(t_l) - \frac{(\tau_f - \tau_i)}{2} \frac{u}{r}\Big|_k &= 0 \\ \sum_{l=0}^{N} D_{kl} w\,(t_l) - \frac{(\tau_f - \tau_i)}{2} \left(\frac{T\sin\beta}{m} - \frac{\mu}{r^2} + \frac{u^2}{r} \right)\Big|_k &= 0 \\ \sum_{l=0}^{N} D_{kl} u\,(t_l) - \frac{(\tau_f - \tau_i)}{2} \left[\frac{T\cos\beta}{m} - \frac{uw}{r} \right]\Big|_k &= 0 \\ \sum_{l=0}^{N} D_{kl} m\,(t_l) - \frac{(\tau_f - \tau_i)}{2} \left(\frac{T}{I_{sp}g} \right)\Big|_k &= 0 \\ k = 0, ..., N \end{aligned} \tag{11}$$

3.4 Rough Braking Phase

The objective of this stage 1 is to break the orbital velocity as much as possible and ensuring the lander is at proper altitude and attitude for camera imaging. This stage starts (s_i) from the terminal condition of the transfer phase explained in subsection 3.1. The terminal altitude of this phase is specified by the camera imaging capability. Camera image resolution improves as altitude decreases, so the sensor team specifies the altitude at which the camera image resolution is better for processing. The orientation of the lander at the terminal condition of this stage (s_f) is specified by sensor team based on camera mounting angle and cameras field of view.

$$\left[\, r_1(t_i) - r_{1s_i} \;\; w_1(t_i) - w_{1s_i} \;\; u_1(t_i) - u_{1s_i} \,\right]^T = 0$$
$$\left[\, r_1(t_f) - r_{1s_f} \;\; \beta_1(t_f) - \beta_{1s_f} \,\right]^T = 0$$

3.5 Attitude Hold Phase

The objective of this stage 2 is to image the lunar surface, process it and feed the navigation information to guidance algorithm for guiding the lander to hazard free landing site. The initial condition of this phase has been specified by the rough braking phase. We note that lander is already in favorable orientation for camera imaging since we enforced it as terminal condition in the former phase. During this phase, the thrust firing and attitude has to be with held to avoid jittering during image capturing.

$$\left[\, T_2(t) - T_{1s_f} \;\; \beta_2(t) - \beta_{1s_f} \,\right]^T = 0$$

3.6 Retargeting Phase

At terminal condition of former phase, the position of the lander with respect to hazard free lunar surface is known. The objective of this stage 3 is to guide the lander from the former phases terminal condition to an altitude of few hundred meters above the landing site. Also at terminal point it has to be ensured zero kinetic energy as well as landers orientation has to be vertical above the site favouring landing.

$$\left[\, r_3(t_f) - r_{3s_f} \;\; \beta_3(t_f) - \beta_{3s_f} \,\right]^T = 0$$
$$\left[\, w_3(t_f) - w_{3s_f} \;\; u_3(t_f) - u_{3s_f} \,\right]^T = 0$$

4. NUMERICAL RESULTS

Simulation analysis has been carried out using optimal control formulation explained earlier. The platform used for simulation is object-oriented reusable software package DIDO. After formulating all constraints, DIDO exploits the problem solving environment of MATLAB and uses the suite of mathematical nonlinear programming solvers.

Here, the problem is formulated as explained in section 3. The problem consists of set of equality and non-equality bounds. Normalization is needed so that the algorithm searches for feasible solution in same non-dimensional domain which has been discussed earlier.

A set of state and control conditions is provided for the algorithm as guess variables though they didnt need to satisfy any constraints. Once the final error is less than the tolerance value specified, the computation stops. If the tolerance value is never met then check whether problem has been defined properly. However, as part of the verification, we can use the control profile to propagate the problem dynamics with small time step by Runge-Kutta 4 and the results have been validated.

4.1 Optimal Control Constraint

Initial and final condition for the point mass dynamics is tabulated in Table 1. In formulation, these condition become constraint for optimal control algorithm. As seen in table, each stage has its own mission constraint. Stage

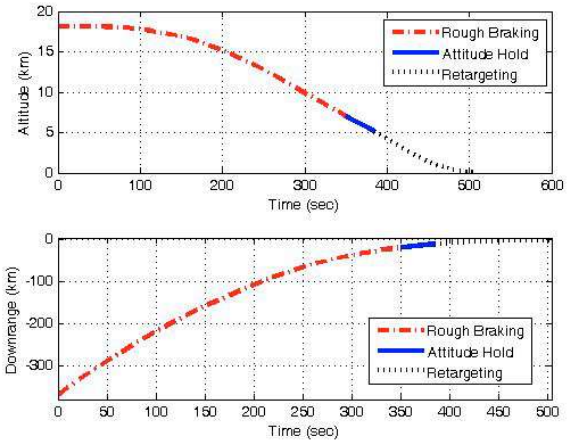


Fig. 1. Altitude Trajectory Implying Descent of the Landercraft

1 terminal condition in altitude and attitude has been specified by the sensor team. Stage 2 constraint emphases on avoiding jittering during imaging. Stage 3 constraints enforces soft vertical landing on landing site. Detailed description of these constraints has been discussed in the section 3.

Table 1. Thrust Constraint for Optimal Control Problem

Stage	Variables	Initial	Final
1	Radial Vel.	0	Free
	Tangent Vel.	1.69 km/s	Free
	Altitude	18.2 km	7 km
	Thrust Orientation	Free	140 deg
2	Radial Vel.	Stage 1 T.C.	Free
	Tangent Vel.	Stage 1 T.C.	Free
	Altitude	Stage 1 T.C.	Free
	Thrust Orientation	140 deg	140 deg
3	Radial Vel.	Stage 2 T.C.	0
	Tangent Vel.	Stage 2 T.C.	0
	Altitude	Stage 2 T.C.	100 m
	Thrust Orientation	Stage 2 T.C.	90 deg

Table 2 implies the thrust control constraint in each of the phases. Table 3 gives the orbital parameters used for the optimal control problem simulation.

Table 2. Thrust Constraint for Optimal Control Problem

Stage	Upper Bound	Lower Bound
1	100% of $800 * 4$ N	100% of $800 * 4$ N
2&3	50% of $800 * 4$ N	50% of $800 * 4$ N

Table 3. Mission Parameter for Optimal Control Problem

Variables	Values
Gravity	1.62 m/s^2
Gravitational parameter	$4902.779E9$ km^3/s^2
Moon radius	1737.1 km

Fig. 1 shows the altitude of the lander decreases satisfying all mission constraint i.e. first stage terminal constraint of 7 km and third stage terminal constraint of 100 m. Fig. 1 also shows the down track traced by the lander craft with crafter reaching desired landing site at final time.

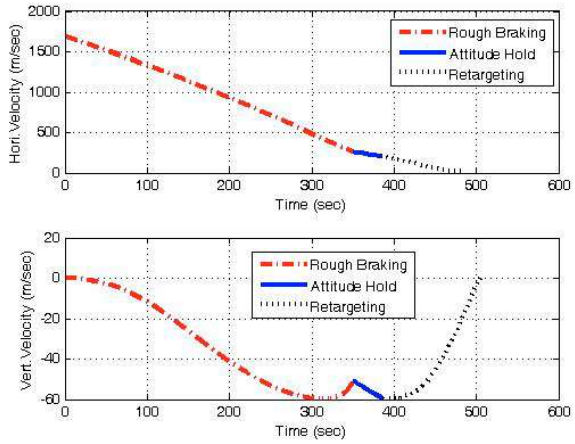


Fig. 2. Tangential Velocity Profile of Landercraft

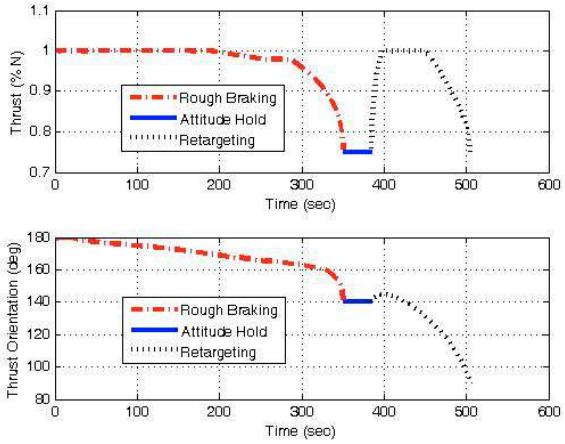


Fig. 3. Demanded Thrust Firing for Achieving Mission Goals

Fig. 2 shows the velocity components of the lander craft reaching the mission constraints of ensuring soft landing. Note the terminal velocity condition achieved is optimal since terminal velocity condition of the first stage is set free. Fig. 3 is the demanded thrust throttling percentage after satisfying the mission constraints given in Table 2.

Fig. 3 depicts the thrust vector direction with terminal constraints satisfied i.e. first stage terminal constraint of 140 *deg* and third stage terminal constraint 90 *deg* making sure of vertical landing at desired site. Note that in stage 2 the throttling percentage and thrust vector direction of former phase is maintained throughout as mission required as given in equation 12.

In LPS method since all dynamic constraints and terminal constraints are converted into equality and inequality constraints, the algorithm can take into account of additional constraints of desired landing site. LPS method is mathematically complex. The number of LGL points determines the dimensional complexity of the problem. If number of points chosen is less, then the solution is not smooth. So optimal number of points is chosen based on experience; no theoretical proof is available. Using pseudospectral knots multistage lunar landing problem is solved satisfying all mission constraint in single optimal control framework.

5. CONCLUSION

Thus multiphase lunar landing problem is solved in optimal control framework using LPS method with pseudospectral knots. In optimal control framework all mission constraint like thrust upper-lower bound, camera look angle constraint and time for image processing are considered. Due to these constraint, landing problem is split into various stages. By simulation the enforcement of terminal mission constraints in each stage has been validated. Along with these constraints, proper orientation of lander favorable for lander leg with zero velocity is satisfied over the landing site.

REFERENCES

Cho, D.H., Jeong, B., Lee, D., and Bang, H. (2009). Optimal perilune altitude of lunar landing trajectory. *International Journal of Aeronautical and Space Sciences*, 10.

Fahroo, F. and Ross, M. (2004). Pseudospectral method knotting methods for solving optimal control problems. *Journal of Guidance, Control, and Dynamics*, 27, 397–405.

Hawkins, A.M. (2005). Constrained trajectory optimization of a soft lunar landing from a parking orbit.

Kozynchenko, A.I. (2010). Enhancing the manoeuvring capabilities of a lunar landing module using predictive guidance algorithms. *Acta Astronautica*, 67, 406–416.

Lee, A.Y. (2011). Optimal terminal descent guidance logic to achieve a soft lunar touchdown. *California Institute of Technology*.

Mathavaraj, S., Pandiyan, R., Ghatpande, N.D., and Gopinath, N. (2012). A predictive guidance scheme for soft landing of a lunar module. *63rd International Astronautical Congress, Italy*.

Park, B.G., Ahn, J.S., and Tahk, M.J. (2011). Two-dimensional trajectory optimization for soft lunar landing considering a landing site. *International Journal of Aeronautical and Space Science*, 12, 288–295.

Rekha, A., Shukkoor, A.A., and Mohanlal, P. (2012). Challenges in navigation system design for lunar soft landing. *Trivandrum Conference, VSSC*.

Vaeth, J. and Sarles, M. (1964). A lunar landing guidance system for soft-precision landings. *International Convention on Space Electronics and telemetry*.

Vinh, N., Busemann, A., and Culp, R. (1980). *Hypersonic and planetary entry flight mechanics*. Univ. of Michigan Press, Ann Arbor.

Zhou, J., Teo, K., Zhou, D., and Zhao, G. (2010). Optimal guidance for lunar module soft landing. *Non-linear dynamics and Systems theory*, 10, 189–201.